1. UVOD

1.1 LASTNA INFO.

SPOROČILO:

KOVANEC → 50/50 } $n \cdot n \cdot n \dots \to n^l$

l KOVANCEV VRŽEMO

$$A(n^l) = l \cdot f(n) \Rightarrow k \cdot \log_b(n)$$

$b = 2$ – BITI
$b = 3$ – TRITI ..

SHANON – RAZLIČNE VRJETNOSTI

$$k \cdot \log(1/p) = \boxed{-k \cdot \log_b(p)}$$

k → POVOLJITVE

p → VRJETNOST

OD ZDES NAPREJ:

$$\log(x) := \log_2(x)$$

LOGARITMI:

$$\log(a \cdot b) = \log(a) + \log(b)$$

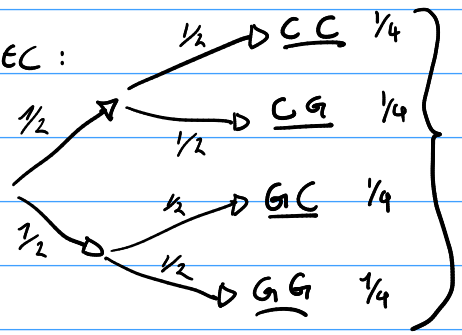
$$\log(1) = 0$$

$$\log(a^b) = b \cdot \log(a)$$

$$\log_b(x) = \frac{\log_a(x)}{\log_a(b)}$$

$$\underbrace{\ln(x)' = \frac{1}{x}}_{\text{ODVOD}} \Rightarrow \ln(z) \leq z - 1$$

KOVANEC:



$$-4 \cdot \log(1/4) = 4 \cdot (-0{,}5) = \underline{\underline{2 \text{ BITA}}}$$

1.2 ENTROPIJA

MERA ZA NEDOLOČENOST SISTEMA

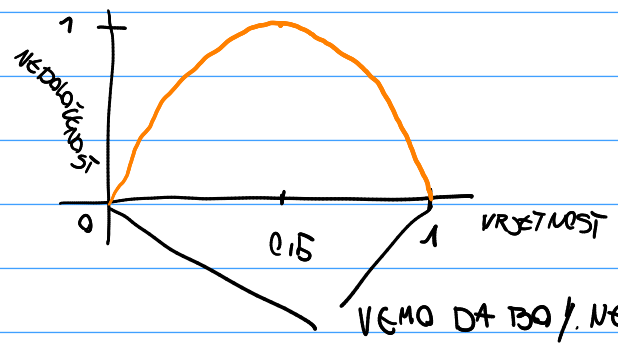
INFO VRJETNOST

$X = \{C, G\}$ $\quad I_C = -\log(p)$ (INFORMACIJA ZA CIFRO)

$P = \{p, 1-p\}$ $\quad I_G = -\log(1-p)$ (INFO. ZA GRB)

$\bar{I} = p_C \cdot I_C + p_G \cdot I_G$

$= -p_C \cdot \log(p_C) + -p_G \cdot \log(p_G)$



VEMO DA BO / NE BO

↳ FUL DOLOČENO

↳ MASHNA NEDOLOČENOST / ENTROPIJA

DISKRETNI VIR /

NAKLJUČNA SPREMENLJIVKA

$X = \{x_1, x_2, x_3, \dots, x_n\}$

$P = \{p_1, p_2, p_3, \dots, p_n\}$

$\boxed{\sum_{i=1}^{n} p_i = 1}$

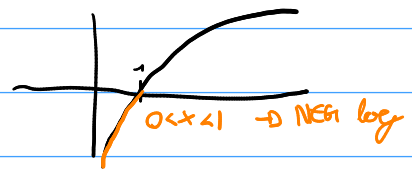
VRJETNOSTI SE SEŠTEJEJO V 1

ENTROPIJA:

$$H(x) = -\sum_{i=1}^{n} p_i \cdot \log(p_i)$$

LASTNOSTI:

- SIMETRIČNA
- ZVEZNA



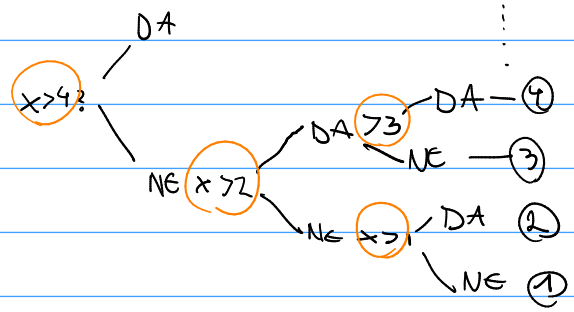
- $H(x) \geq 0 \Leftarrow p_i \geq 0 \Rightarrow -p_i \log(p_i) \geq 0$
- ZGORAJ OMEJENA Z $\log(n)$

  ↳ DOKAZ ☺
- ADITIVNA x, y $\quad H(x,y) = H(x) + H(y)$; x, y NEODVISNA

PRIMERI: UGIBAMO ŠTEVILA OD 1 DO 8

$x = \{[1:8]\}$

$p = \{1/8, 1/8 \dots\}$ $\quad H(x) = -\sum_{i=1}^{8} \frac{1}{8} \cdot \log(1/8) = \log(8) = \underline{3 \text{ BiTi}}$



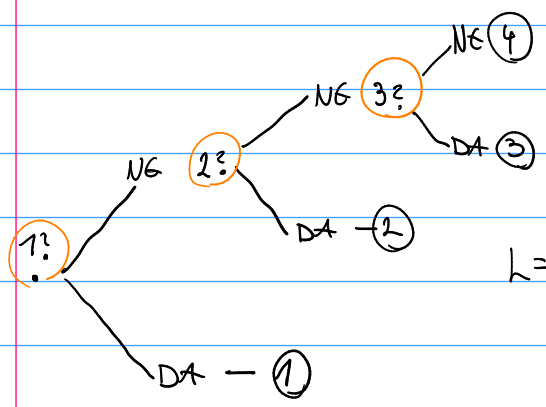
ZMERI 4 VPRAŠANJA

b) $X = \{1, 2, 3, 4\}$ $\quad H(x) = \sum_{i=1}^{4} p_i \log(p_i)$

$p = \{1/2, 1/4, 1/8, 1/8\}$

$= -\frac{1}{2}\underbrace{\log(1/2)}_{-1} - \frac{1}{4}\underbrace{\log(1/4)}_{-2} - \frac{1}{8}\underbrace{\log(1/8)}_{-3} \cdot 2$

$= \underline{\underline{\frac{7}{4}}}$ BITOV

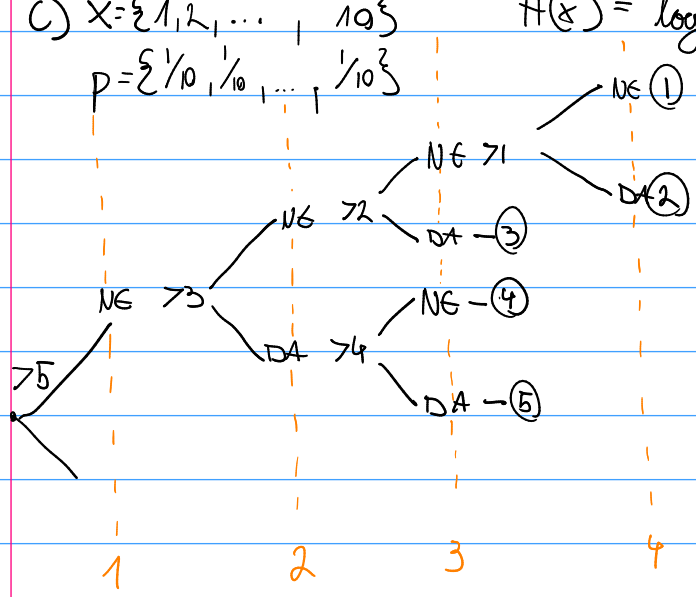


$L = \frac{1}{2} \cdot 1 + \frac{1}{4} \cdot 2 + \frac{1}{8} \cdot 3 = \underline{\underline{\frac{7}{4}}}$

c) $X = \{1, 2, \dots, 10\}$ $\quad H(x) = \log_2 10 = \underline{\underline{3{,}32}}$ ⇝ BEST CASE! SPODNJA MEJA

$p = \{1/10, 1/10, \dots, 1/10\}$

NI ISTO



$L = \frac{6}{10} \cdot 3 + \frac{4}{10} \cdot 4 = \underline{\underline{3.4}}$

NE DOSEŽEMO SP MEJE